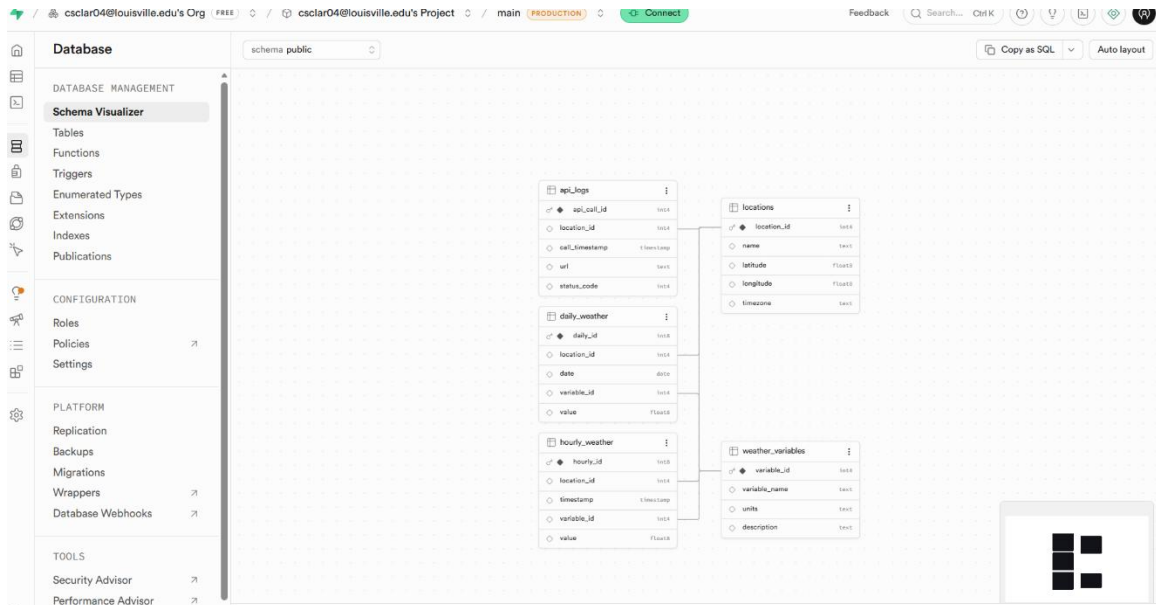


Database/ Storage Documentation

MSBA 962- Summer 2026

ERD in Supabase



ER Diagram Description

- **locations**
 - 1 → many hourly_weather
 - 1 → many daily_weather
 - 1 → many api_logs
- **weather_variables**
 - 1 → many hourly_weather
 - 1 → many daily_weather
- **hourly_weather**
 - many → 1 locations
 - many → 1 weather_variables
- **daily_weather**
 - many → 1 locations
 - many → 1 weather_variables

- **api_logs**
 - many → 1 locations

Database Schema Documentation

Weather Data Engineering Pipeline

Storage Layer: SQLite or CSV

Source: Open-Meteo API

Purpose: Store hourly and daily weather data for ETL and dashboard visualization.

Table: locations

Stores metadata for each city or coordinate queried.

COLUMN	TYPE	DESCRIPTION
LOCATION_ID (PK)	SERIAL	Unique ID
NAME	TEXT	City or label
LATITUDE	DOUBLE PRECISION	Latitude
LONGITUDE	DOUBLE PRECISION	Longitude
TIMEZONE	TEXT	Timezone returned by API

Table: weather_variables

Stores the weather metrics you track.

Column	Type	Description
variable_id (PK)	SERIAL	Unique ID
variable_name	TEXT	API variable name
units	TEXT	Units (° F, %, mm, mph)
description	TEXT	Human-readable description

Table: hourly_weather

Stores hourly weather values in long format.

Column	Type	Description
hourly_id (PK)	BIGSERIAL	Unique ID
location_id (FK)	INT	References locations
timestamp	TIMESTAMP	Hour timestamp
variable_id (FK)	INT	References weather_variables
value	DOUBLE PRECISION	Weather value

Table: daily_weather

Stores daily aggregated values.

Column	Type	Description
daily_id (PK)	BIGSERIAL	Unique ID
location_id (FK)	INT	References locations
date	DATE	Calendar date
variable_id (FK)	INT	References weather_variables
value	DOUBLE PRECISION	Daily aggregated value

Table: api_logs

Tracks API calls for reproducibility.

Column	Type	Description
api_call_id (PK)	SERIAL	Unique ID
location_id (FK)	INT	References locations
call_timestamp	TIMESTAMP	When the API was called
url	TEXT	Full API URL
status_code	INT	HTTP status

3. Initial Load Script in Supabase SQL editor

 SQL

Insert Louisville Location and Weather Variables

+

```
1 INSERT INTO locations (name, latitude, longitude, timezone)
2 VALUES ('Louisville, KY', 38.2527, -85.7585, 'America/New_York');
3
4 INSERT INTO weather_variables (variable_name, units, description)
5 VALUES
6 ('temperature_2m', '°F', 'Air temperature at 2 meters'),
7 ('relative_humidity_2m', '%', 'Relative humidity at 2 meters'),
8 ('precipitation', 'mm', 'Hourly precipitation'),
9 ('wind_speed_10m', 'mph', 'Wind speed at 10 meters');
10
```

Results

Explain

Chart

Export ▾

Success. No rows returned